



YEAR 12

ASSESSMENT TEST 2
TERM 1, 2013

MATHEMATICS

EXTENSION 2

Time Allowed – 90 Minutes
(Plus 5 minutes Reading Time)

General Instructions:

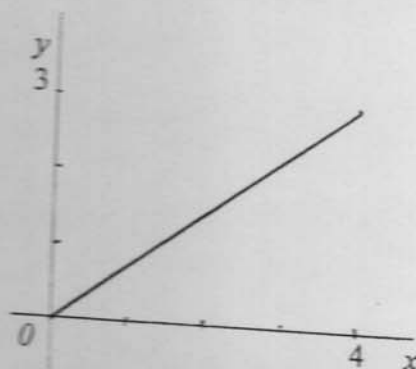
- All questions may be attempted
- All questions are of equal value
- Standard Integral Tables will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

The answers to all questions are to be returned in separate bundles clearly labelled Question 1, Question 2, etc.

Each question must show your (in the top right hand corner) Candidate Number.

Question 1.

- | | Marks |
|--|-------|
| (a) Using integral tables find the primitive function of $\frac{1}{\sqrt{x^2+4}}$. | 1 |
| (b) Find $\int \frac{\sin(x+\frac{\pi}{4})}{\sin x} dx$ | 2 |
| (c) Neatly graph the conic with equation : $\frac{x^2}{6} + \frac{y^2}{4} = 1$, showing all relevant features.
Use a scale of 1 unit = 1 cm. | 5 |
| (d) Find the Cartesian equation for a conic with the Origin as centre,
a focus located at (5,0) and directrix with equation $x = 7$. | 2 |
| (e) The graph of $y=f(x)$ for the interval $[0, 4]$ is shown below. | 2 |



On the same axes graph $y = f(x)$, $y = f(8-x)$ and $y = f(x-8)$.
Use a scale of 1 unit = 1 cm.

A conic has parametric equations $x = \frac{e^{3t}+e^{-3t}}{2}$ and $y = \frac{e^{3t}-e^{-3t}}{4}$.

- (i) Find the Cartesian equation
- (ii) Find the eccentricity of the conic

2
1

Question 2.

Prove $\int_0^1 x^3(1-x)^{11} dx = \int_0^1 x^{11}(1-x)^3 dx$.

(i) Find the values a , b and c such that $\frac{ax+b}{x^2+2x+2} - \frac{ax+c}{x^2-2x+2} = \frac{1}{x^4+4}$

ii) Hence find $\int \frac{dx}{x^4+4}$.

Graph $y = 2^{\frac{1}{x}}$ showing all asymptotes and intercepts with the axes.
Use a scale of 1 unit = 1 cm.

Find $\int \sqrt{\frac{4+x}{3-x}} dx$

2
2
4
4
3

Question 3.

Find $\int e^{\sqrt{x}} dx$.

Marks

4

The graph of $y=f(x)$ is shown on the attached sheet.
Graph the relation $y^2=f(x)$ on the attached sheet.

4

Let $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x dx$ where n is a positive integer.

(i) Using integration by parts show $I_n = \frac{2^{n-2}\sqrt{3}}{n-1} + \frac{n-2}{n-1} I_{n-2}$

4

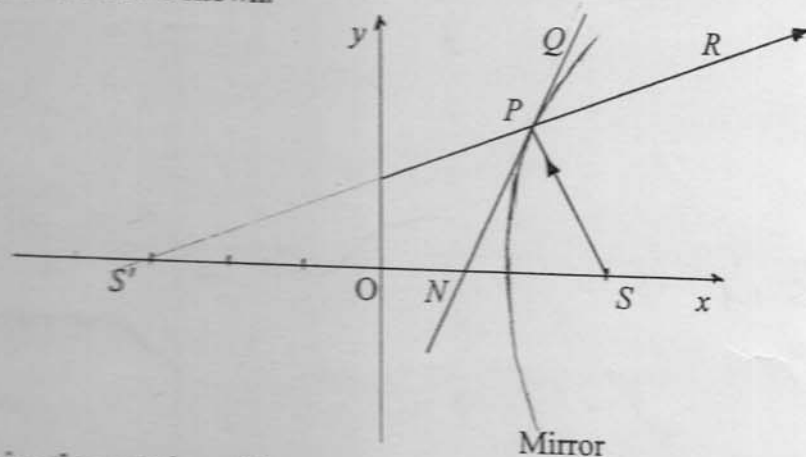
(ii) Evaluate I_3

3

Question 4.

A hyperbolic mirror with equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and foci $S(ae, 0)$ and $S'(-ae, 0)$ is shown below.

A ray of light emanates from the focus $S(ae, 0)$, and is reflected from the mirror at the point $P(x_1, y_1)$ towards R is shown.



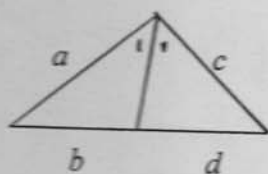
(i) Derive the equation of the tangent at $P(x_1, y_1)$.

2

(ii) The tangent at P intersects the x axis at N .
Prove that the tangent at P bisects $\angle SPS'$.
Assume the angle bisection result:

4

$$\frac{a}{b} = \frac{c}{d}$$



(iii) Given that the angle of incidence equals the angle of reflection ($\angle SPN = \angle QPR$) show that the reflected ray appears to emanate from the other focus $S'(-ae, 0)$.

2

(i) Show that $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{4+3\cos x} = \frac{4}{\sqrt{7}} \tan^{-1}\left(\frac{1}{\sqrt{7}}\right)$

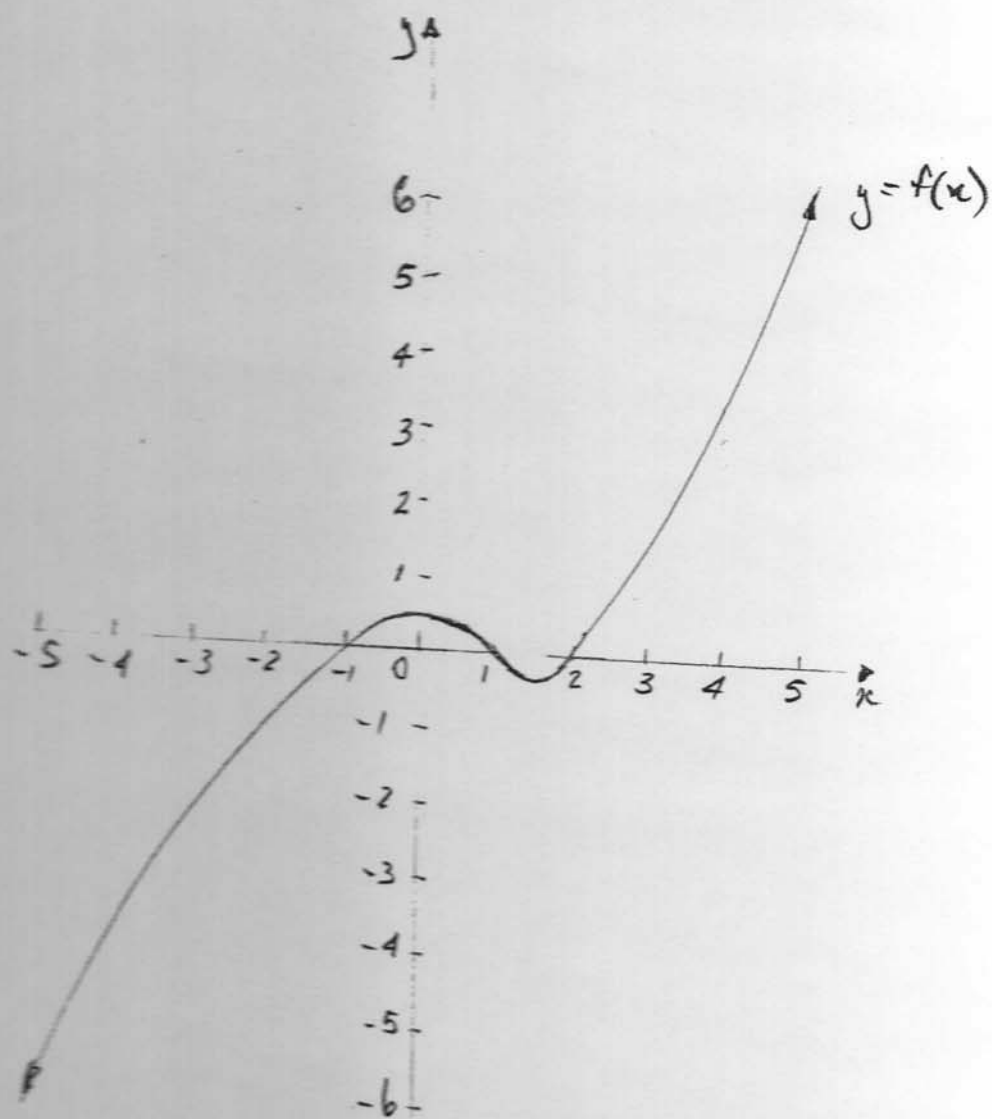
3

(ii) Using (i) find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos 2x dx}{4+3\cos x}$

4

End Of Exam

3(d)



MATHEMATICS Extension 2: Question...!

Suggested Solutions	Marks	Marker's Comments
(a) $\int \frac{dx}{\sqrt{x^2+4}} = \ln(x + \sqrt{x^2+4}) + C$ from integral table	①	①
(b) $\int \frac{\sin(x+\pi/4)}{\sin x} dx = \int \frac{(\sin x \cos \pi/4 + \cos x \sin \pi/4)}{\sin x} dx$ $= \int \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \frac{\cos x}{\sin x} \right) dx$ $= \frac{1}{\sqrt{2}} x + \frac{1}{\sqrt{2}} \ln \sin x + C$	②	① expansion ① answer.
(c) $\frac{x^2}{6} + \frac{y^2}{4} = 1$ $a^2 = 6, b^2 = 4, b^2 = a^2(1-e^2)$ $e^2 = 1 - \frac{4}{6}, a = \sqrt{6}, a > 0$ $b = 2, b > 0$ $e^2 = \frac{1}{3}, e = \frac{1}{\sqrt{3}}, 0 < e < 1$ foci $S(a, 0) = (\sqrt{6}, 0), S'(-a, 0) = (-\sqrt{6}, 0)$ directrices $x = \pm \frac{a}{e}, x = \pm 3\sqrt{2}$ $x = -3\sqrt{2}$ $x = 3\sqrt{2}$	⑤	① directrices ① foci ① $\pm a$ ① $\pm b$ ① shape/ scale/axes.
(d) $ae = 5, a = 7, \therefore a^2 = 35$ $a = \sqrt{35}, a > 0$ $\therefore e = \frac{5}{\sqrt{35}}, 0 < e < 1$ $b^2 = a^2(1-e^2), b^2 = 35(1 - \frac{25}{35})$ $\therefore b^2 = 10, x^2 + \frac{y^2}{10} = 1$	②	① e or e^2 must write $a > 0$ or $e > 0$ if a or e found. ① answer.
(e)	②	① $f(x-8)$ ① $f(8-x)$

MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
(P) (i) $x = \frac{e^{3t} + e^{-3t}}{2}$ $y = \frac{e^{3t} - e^{-3t}}{2}$ As it is a conic consider x^2 and y^2. $(2x)^2 = e^{6t} + 2 + e^{-6t}$ $(2y)^2 = e^{6t} - 2 + e^{-6t}$ $4x^2 - 4y^2 = 4$ $\therefore x^2 - y^2 = 1$ $\therefore a^2 = 1$ $b^2 = 1/4$ $\therefore$ hyperbola $\therefore e > 1$ (ii) $b^2 = a^2(e^2 - 1)$ $\frac{1}{4} = 1(e^2 - 1)$ $e^2 = \frac{5}{4}$ $e = \frac{\sqrt{5}}{2}$ $e > 1$	(3)	① $\left. \begin{matrix} x^2 \\ y^2 \end{matrix} \right\}$ ① formula ① 'e' value

Suggested Solutions

Marks

Marker's Comments

a) Let $u = 1 - x$ $du = -dx$
 $x = 0, u = 1$; $x = 1, u = 0$

$$\int_0^1 x^3(1-x)^4 dx = -\int_1^0 (1-u)^3 u^4 du$$

$$= \int_0^1 u^4(1-u)^3 du = \int_0^1 x^4(1-x)^3 dx$$

(dummy variable)

b) $(x^2 + 2 - 2x)(x^2 + 2 + 2x) = (x^2 + 2)^2 - 4x^2$
 $= x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$

$$(ax + b)(x^2 - 2x + 2) - (ax + c)(x^2 + 2x + 2) = 1$$

Sub $x = 0$ $b - c = \frac{1}{2}$ (1)

$x = 1$ $-4a + b - 5c = 1$ (2)

$x = -1$ $5b - 4a - c = 1$ (3)

Solve (2) (3) simultaneously. $a + c = -\frac{1}{8}$ (4)

$c - a = -\frac{3}{8}$ (5)

Solve (4), (5) simultaneously

$a = \frac{1}{8}$ $c = -\frac{1}{4}$

$\therefore b = \frac{1}{4}$

$$\int \frac{dx}{x^4 + 4} = \int \frac{\frac{1}{8}x + \frac{1}{4}}{x^2 + 2x + 2} - \int \frac{\frac{1}{8}x - \frac{1}{4}}{x^2 + 2x + 2}$$

$$= \frac{1}{16} \int \frac{2x + 2}{x^2 + 2x + 2} + \frac{1}{8} \int \frac{dx}{(x+1)^2 + 1}$$

$$- \frac{1}{16} \int \frac{2x - 2}{x^2 + 2x + 2} + \frac{1}{8} \int \frac{dx}{(x-1)^2 + 1}$$

well done

some use property

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$\frac{1}{2}$

any relevant equation $\frac{1}{2}m$

but if none of the equations are right won't get any mark here

$\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}m$ for a, b, c (a)

if a = 0, max 1m out of 3 marks

most student got this

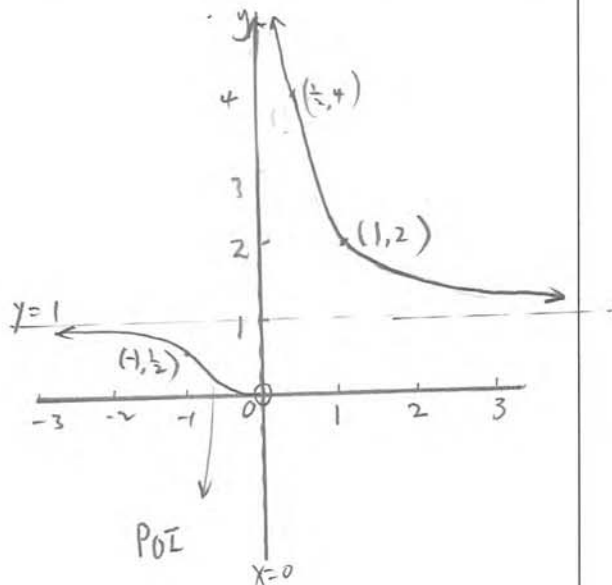
Suggested Solutions

Marks

Marker's Comments

$$= \frac{1}{16} \ln \left(\frac{x^2 + 2x + 2}{x^2 - 2x + 2} \right) + \frac{1}{8} \left(\tan^{-1}(x+1) + \frac{1}{8} \tan^{-1}(x-1) \right) + c$$

c) $y = 2^{\frac{1}{x}}$



$$\begin{aligned} \text{d) } \int \sqrt{\frac{4+x}{3-x}} dx &= \int \frac{4+x}{\sqrt{(4+x)(3-x)}} dx \\ &= \int \frac{4+x}{\sqrt{12-x-x^2}} dx \end{aligned}$$

$$\begin{aligned} &= -\frac{1}{2} \int \frac{-2x-1}{\sqrt{12-x-x^2}} dx + \frac{7}{2} \int \frac{dx}{\sqrt{\frac{49}{4} - (x+\frac{1}{2})^2}} \\ &= -\frac{2}{2} \sqrt{12-x-x^2} + \frac{7}{2} \sin^{-1} \left(\frac{2x+1}{7} \right) + c \\ &= -\sqrt{12-x-x^2} + \frac{7}{2} \sin^{-1} \left(\frac{2x+1}{7} \right) + c \end{aligned}$$

$$1 + \frac{1}{2} + \frac{1}{2}$$

$\frac{1}{2}$ m for each

Asymptotes $x=0$
 $y=1$

Pts $(1, 2)$ $(-1, \frac{1}{2})$

Right branch goes to
Left branch asymptotes

1 m for POI

Very few students
get POI.

many forgot to
number their scales
on the axes

POI $\hat{=}$ $(-0.3466, 0.1353)$

1 m

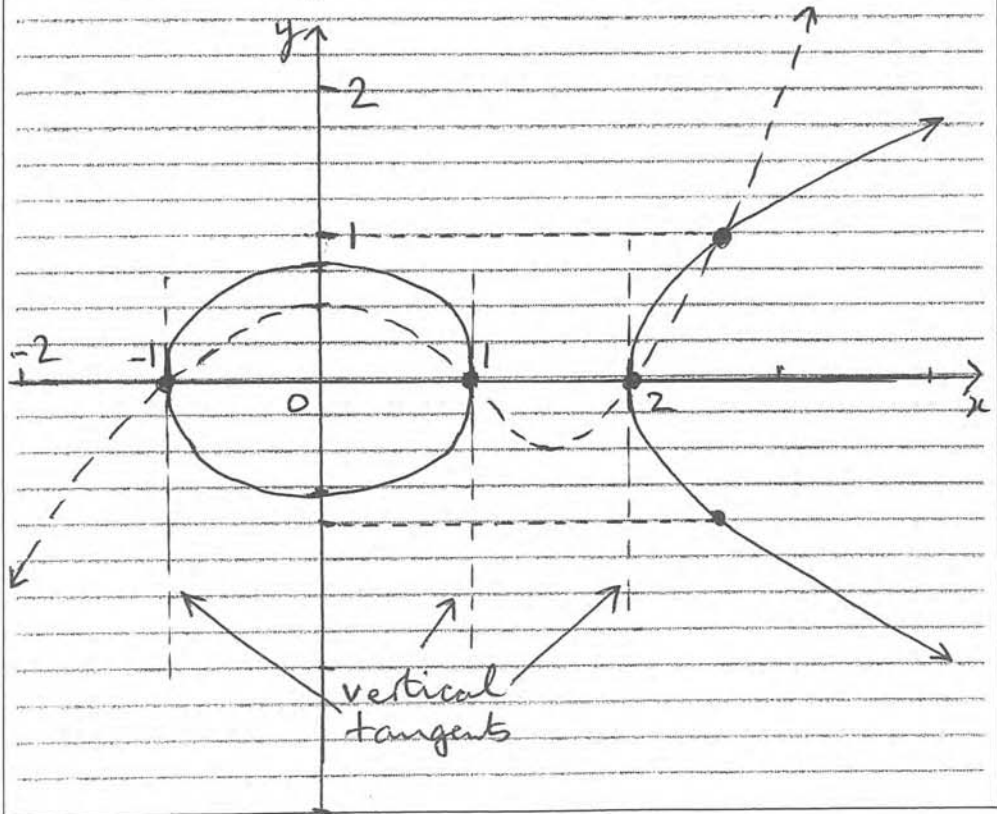
$\frac{1}{2} + \frac{1}{2}$

must show

$$-\frac{1}{2} \int \frac{-2x-1}{\sqrt{12-x-x^2}} dx$$

(many fudging
answers)

$\frac{1}{2} + \frac{1}{2}$

Suggested Solutions	Marks	Marker's Comments
a) $I = \int e^{\sqrt{x}} dx$ Let $u^2 = x$ ($u > 0$) "2u du = dx" $= \int 2ue^u du$ $= 2ue^u - \int 2e^u du$ (By Parts) $= 2ue^u - 2e^u + k$ $= \underline{\underline{2(\sqrt{x}-1)e^{\sqrt{x}} + k}}$	1 1 1 1	Correct substitution Get to this form Substitute back ($\frac{1}{2}$ off for no 'k')
b) Draw $y = \sqrt{f(x)}$ and reflect in $y=0$ for $y^2 = f(x)$ Domain $f(x) \geq 0 \quad \therefore \{-1 \leq x \leq 1\} \cup \{x: x \geq 2\}$ Invariant points $f(x) = \sqrt{f(x)} \Rightarrow f(x) = 0$ or 1 $\frac{dy}{dx} = \frac{f'(x)}{2\sqrt{f(x)}} \Rightarrow 0$ where $f'(x) = 0$ ∞ where $f(x) = 0$ 	1 $\frac{1}{2}$ 1 $\frac{1}{2}$ 1	Symmetry in $y=0$ Implied domain Invariant points for same max/min but larger absolute values verticals at $x=2$ and $x=\pm 1$. Vertical curve either had to be very obvious or specifically mentioned.

Suggested Solutions	Marks	Marker's Comments
c) i) $I_n = \int_{\pi/6}^{\pi/2} \csc^2 x \csc^{n-2} x \, dx$ $= \left[-\cot x \csc^{n-2} x \right]_{\pi/6}^{\pi/2}$ $- (n-2) \int_{\pi/6}^{\pi/2} (-\cot x \csc^{n-3} x) (-\cot x \csc x) \, dx$ $= -0 - (-\sqrt{3} \cdot 2^{n-2}) - (n-2) \int_{\pi/6}^{\pi/2} \cot^2 x \csc^{n-2} x \, dx$ $= \sqrt{3} 2^{n-2} - (n-2) \int_{\pi/6}^{\pi/2} (\csc^2 x - 1) \csc^{n-2} x \, dx$ $= \sqrt{3} 2^{n-2} - (n-2) \int_{\pi/6}^{\pi/2} \csc^n x - \csc^{n-2} x \, dx$ $I_n = \sqrt{3} 2^{n-2} - (n-2) I_n + (n-2) I_{n-2}$ $\therefore (n-1) I_n = \sqrt{3} 2^{n-2} + (n-2) I_{n-2}$ $I_n = \frac{\sqrt{3} 2^{n-2}}{(n-1)} + \frac{(n-2)}{(n-1)} I_{n-2}$	1 1/2 1 1/2 1	(Integration by parts) numerical substitution use of $\cot^2 x = \csc^2 x - 1$ Revert to the I_k forms Rearrangement to final form [Much fudging of signs thro' the integrals]
ii) $I_3 = \frac{\sqrt{3} \times 2^1}{2} + \frac{1}{2} I_1$ $I_3 = \sqrt{3} + \frac{I_1}{2}$ But $I_1 = \int_{\pi/6}^{\pi/2} \csc x \, dx$ $= \left[-\ln(\cot x + \csc x) \right]_{\pi/6}^{\pi/2}$ $= -\ln(1) + \ln(2 + \sqrt{3})$ $\therefore I_3 = \sqrt{3} + \frac{1}{2} \ln(2 + \sqrt{3})$	1 1 1/2 1/2	Simple formula Correct integral Correct substitution

Question 4

For tangent at P

$$\frac{2x}{a^2} - \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\therefore \text{Gradient of Tangent at P} = \frac{b^2 x_1}{a^2 y_1}$$

1

$$\text{Tangent at P } y - y_1 = \frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y_1 y - a^2 y_1 y_1 = b^2 x_1 x - b^2 x_1 x_1 \quad (\div a^2 b^2)$$

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$$

$\frac{1}{2}$

Since P(x_1, y_1) lies on hyperbola

$$\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$$

Some students failed to explain where the = 1 came from. On the whole well done

$$\therefore \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \text{ equation of tangent}$$

$\frac{1}{2}$

(ii) Coordinates of N ($\frac{a^2}{x_1}, 0$)

$$NS = ae - \frac{a^2}{x_1}$$

1

$$= \frac{a}{x_1} (ex_1 - a)$$

$$NS' = ae + \frac{a^2}{x_1}$$

$$= \frac{a}{x_1} (ex_1 + a)$$

By definition of hyperbola

$$\therefore PS = ePM \quad \therefore PS' = ePM'$$

1

$$PS = e \left(x_1 - \frac{a}{e} \right)$$

$$= ex_1 - a$$

$PS' = e \left(x_1 + \frac{a}{e} \right)$ $= e x_1 + a$ $\frac{PS}{NS} = \frac{e x_1 - a}{\frac{a}{x_1}(e x_1 - a)}$ $= \frac{x_1}{a}$ $\frac{PS'}{NS'} = \frac{e x_1 + a}{\frac{a}{x_1}(e x_1 + a)}$ $= \frac{x_1}{a}$ $\therefore \frac{PS}{NS} = \frac{PS'}{NS'}$ $\angle SPN = \angle S'PN$ (Angle Bisection Theorem) $\therefore$ The Tangent at P bisects the angle S'PS (iii) $\angle SPN = \angle S'PN$ (from part (i)) $\angle S'PN = \angle QPR$ (given) S'PR is a straight line (vertically opposite angles are equal) $\therefore$ Pr appears to emanate from S' (b) $t = \tan \frac{x}{2} \rightarrow \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$ $\frac{dt}{dx} = \frac{1}{2} (1 + \tan^2 \frac{x}{2})$ $= \frac{1+t^2}{2}$ $dx = \frac{2}{1+t^2} dt$ When $x = \frac{\pi}{2}$ $t = 1$ $x = \frac{-\pi}{2} \quad t = -1$	1 1 1 1	This may be done differently and equivalent marks given Some students proved the angle bisection theorem not what was asked. Those who did not use PS=ePM ended up on most occasions with a messy algebraic expression A number students assumed straight line and stated angles were vertically opposite
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$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{4+3\cos x} = \int_{-1}^1 \frac{\frac{2}{1+t^2} dt}{4+3\frac{1-t^2}{1+t^2}}$$

$$= \int_{-1}^1 \frac{2dt}{4+4t^2+3-3t^2}$$

$$= \int_{-1}^1 \frac{2dt}{t^2+7}$$

$$= \left[\frac{2}{\sqrt{7}} (\tan^{-1} \frac{t}{\sqrt{7}}) \right]_{-1}^1$$

$$= \frac{2}{\sqrt{7}} (\tan^{-1} \frac{1}{\sqrt{7}}) - \frac{2}{\sqrt{7}} (\tan^{-1} \frac{-1}{\sqrt{7}})$$

$$= \frac{2}{\sqrt{7}} (\tan^{-1} \frac{1}{\sqrt{7}}) + \frac{2}{\sqrt{7}} (\tan^{-1} \frac{1}{\sqrt{7}})$$

$$= \frac{4}{\sqrt{7}} (\tan^{-1} \frac{1}{\sqrt{7}})$$

$$(ii) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos 2x dx}{4+3\cos x} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2\cos^2 x - 1 dx}{4+3\cos x}$$

Using part (i)

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2\cos^2 x dx}{4+3\cos x} - \frac{4}{\sqrt{7}} (\tan^{-1} \frac{1}{\sqrt{7}})$$

1

$\frac{1}{2}$

Must have this step or earlier stated $\frac{dx}{4+3\cos x}$ was even function

$\frac{1}{2}$

$\frac{1}{2}$ if just did first step

The use of integration by parts was unsuccessful for this question

1

Very few got this result
many students ran out of
time or used unsuccessful
methods

Divide through $4+3\cos x$

$$\begin{array}{r} \frac{2}{3}\cos x - \frac{8}{9} \\ 4+3\cos x \overline{) 2\cos^2 x} \\ \underline{-} \\ 2\cos^2 x + \frac{8}{3}\cos x \\ \underline{-} \\ -\frac{8}{3}\cos x \\ -\frac{8}{3}\cos x - \frac{32}{9} \\ \underline{-} \phantom{-\frac{8}{3}\cos x} \\ \frac{32}{9} \end{array}$$

1

1

1

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2\cos^2 x - 1}{4 + 3\cos x} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{3}\cos x - \frac{8}{9} + \frac{\frac{32}{9} - 1}{4 + 3\cos x} dx$$

$$= \left[\frac{2}{3}\sin x - \frac{8}{9}x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{23}{9} \times \frac{4}{\sqrt{7}} \left(\tan^{-1} \frac{1}{\sqrt{7}} \right)$$

$$= \frac{4}{3} - \frac{8\pi}{9} + \frac{92}{9\sqrt{7}} \left(\tan^{-1} \frac{1}{\sqrt{7}} \right)$$